

RoboPainting: Indirect Force Control for Stable Brush–Wall Contact

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github.com/chiawenwang/robot_painting

Abstract—This report presents RoboPainting, a contact-rich robotic painting system designed to maintain stable brush–wall contact while following prescribed letter trajectories. The project focuses on feedback controller design for robotic contact rather than mechanical design alone. In simulation, a MuJoCo model of a six-degree-of-freedom Piper manipulator regulates brush normal force through an outer-loop admittance controller that maps force error into a wall-normal position command, which is then converted to joint references through damped inverse kinematics. The simulated controller maintained contact throughout a 6.00 mm wall-retreat event and achieved a mean pen-down tangential tracking error of 0.497 mm. On the real Piper platform, a joint-level MIT-mode impedance controller is used for safer compliant contact. The hardware task was implemented as reverse painting: the wall was colored first, and an eraser mounted at the end effector removed color along the commanded path to write “UCLA.” The recommended hardware stiffness scale is $s = 0.3$, because $s = 0.1$ was too soft for reliable letter formation, whereas $s \geq 0.5$ made the arm too stiff for safe contact. The final recommended architecture uses simulation to tune force-regulation behavior and hardware joint impedance to preserve physical compliance during real-world contact.

Keywords—robotic painting, indirect impedance control, contact-force regulation.

I. INTRODUCTION AND LITERATURE REVIEW

Robotic painting is a representative contact-rich manipulation problem because successful execution requires both geometric path tracking and stable physical interaction. The robot must move the tool tangentially to reproduce a desired pattern while regulating normal interaction so that contact remains continuous without excessive compression. Similar requirements occur in polishing, sanding, grinding, wiping, and other surface-treatment operations. Pure position control is vulnerable to surface uncertainty: inward surface motion can create excessive contact force, whereas outward motion can produce separation and incomplete strokes.

Contact-rich manipulation broadly requires robots to reason about both motion and contact forces. Suomalainen, Karayiannidis, and Kyrki survey robotic manipulation in contact and show that reliable task execution often depends on controlling the interaction with the environment, not only the free-space trajectory [1]. Their survey motivates this project by placing painting in a broader class of tasks where the environment is part of the control loop. Unlike many manipulation examples in that survey, this project studies a visually simple but physically challenging surface-contact



Fig. 1: RoboPainting hardware demonstration. The mobile platform carries a Piper manipulator that maintains contact with a vertical writing surface while an eraser-style end effector traces the UCLA letters. This teaser summarizes the central goal of the project: stable wall-normal contact together with accurate tangential path tracking.

task: writing letters on a wall while maintaining stable brush or eraser contact.

Bansal *et al.* study robotic painting with human-like applicator behavior and show that painting quality is strongly influenced by the interaction between the applicator and the surface [2]. Their work is similar in application, because both projects use robotic actuation to reproduce painting motions. However, their emphasis is on coating outcomes and human-applicator imitation, whereas this project emphasizes feedback-controller design and gain selection for brush–wall contact.

Hogan’s impedance-control framework established the principle of regulating the dynamic relationship between motion and interaction force rather than commanding position or force independently [3]. This idea directly supports the simulation design: measured force error is passed through a virtual mass–damper–spring model before it becomes a wall-normal motion command. Abu-Dakka and Saveriano review variable impedance control and learning approaches for robots that must adapt compliance in changing environments [4]. The hardware implementation in this project is simpler

TABLE I: Controller design requirements and performance metrics.

Design objective	Requirement	Performance metric
Normal force regulation	Desired force = 10 N	Mean force, peak force, and force RMSE during pen-down motion
Contact maintenance	No separation during the 6 mm retreat	Percentage of retreat samples with nonpositive contact gap
Wall following	Brush responds to wall retreat	Brush-front displacement and displacement-following ratio
Tangential path accuracy	Mean < 1 mm; maximum < 2 mm	Euclidean y - z tracking error
Orientation stability	Brush-axis drift < 1°	Maximum tool-axis angular deviation
Transient stability	No sustained oscillation	Force time history and settling behavior

than the learning-based methods in that review, but it shares the same motivation: the robot should be compliant enough to tolerate uncertain contact.

Classic force-control literature further motivates the distinction between direct force control and indirect force/compliance control. Whitney’s historical review identifies stability as a central issue in robot force control [5], and Salisbury’s Cartesian stiffness control demonstrates that apparent end-effector stiffness can be shaped for contact tasks [6]. The present project applies these ideas pragmatically: the simulation uses a wall-normal admittance model, while the real robot uses low joint stiffness in the MIT interface because high-stiffness position control was not appropriate for safe contact.

II. OBJECTIVES AND CONTROLLER DESIGN REQUIREMENTS

The study objective was to determine whether a position-actuated manipulator could maintain contact with a moving surface while preserving a prescribed painting path. The wall was commanded to move forward and backward in the x direction to represent local surface-height variation. The controller was required to regulate a nominal normal force of 10 N, maintain contact throughout a 6 mm wall-retreat event, and avoid sustained force oscillation after contact transitions.

Geometric and orientation requirements were defined to ensure that normal compliance did not degrade the painted shape. During pen-down motion, the mean Euclidean y - z tracking error was required to remain below 1 mm and the maximum error below 2 mm. The brush-axis orientation error was required to remain below 1° . These numerical thresholds are engineering targets for this simulation rather than universal painting standards.

A later hardware study should revise these thresholds according to tool dimensions, surface material, sensor resolution, actuator bandwidth, and the process-quality requirements of the intended application. In the present hardware demonstration, no external force/torque sensor was used;

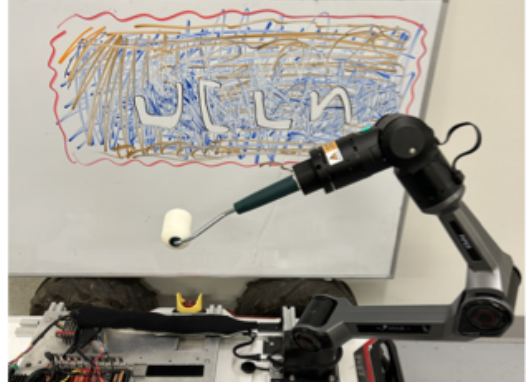


Fig. 2: Close-up of the Piper manipulator during reverse painting. The eraser-style tip is mounted to the end effector so that tangential motion writes the letters while compliant normal motion maintains contact with the wall.

therefore, hardware evaluation focuses on contact stability, safety, visible stroke quality, and the effect of joint-impedance stiffness.

III. ROBOT DESIGN AND SYSTEM OVERVIEW

The system consists of a six-degree-of-freedom AgileX Piper manipulator mounted on a mobile base and carrying a custom eraser-style end effector. The task is to write the word “UCLA” on a vertical whiteboard or wall while maintaining stable physical contact. Fig. 1 shows the complete hardware platform and final visual outcome, and Fig. 2 shows the end-effector contact configuration used during reverse painting.

The wall-normal direction is defined as the x axis, and the painting plane is the wall-tangential y - z plane. The controller intentionally separates these directions. Compliance is primarily introduced in the wall-normal x direction, while the letter trajectory is specified in the y - z plane. This separation is useful because painting quality depends on both letter geometry and stable tool pressure.

In hardware, the final demonstration used reverse painting rather than depositing paint. The wall was first covered with color, and the eraser attached to the end effector removed the color along the commanded path. This produced a negative-space “painting” effect while preserving the same contact-rich control problem as brush painting: the robot still had to maintain controlled normal interaction while tracking a tangential path.

IV. METHODS

A. Simulation Model and Task Definition

The robot was simulated in MuJoCo using a six-degree-of-freedom Piper model adapted from open-source AgileX Robotics model resources [7]. MuJoCo is designed for model-based control and contact-rich articulated simulation [8]. A virtual planar wall was represented as a moving body whose prescribed displacement was along the global x direction. The wall first moved toward the robot and then retreated, providing a repeatable approximation of a locally

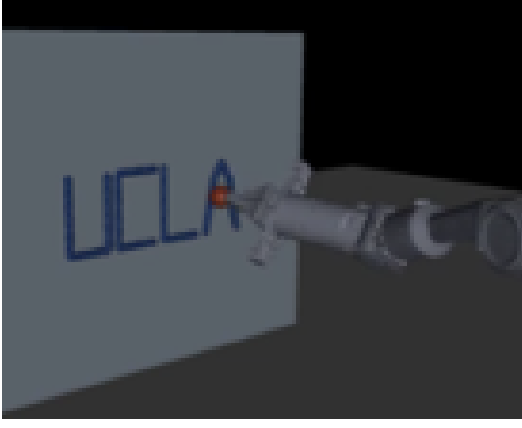


Fig. 3: MuJoCo visualization of the Piper manipulator writing the prescribed UCLA path on the moving wall.

uneven surface and enabling evaluation of both inward-disturbance force response and outward-disturbance contact maintenance.

The letters were represented as piecewise-linear trajectories in the wall-tangential y - z plane. Between disconnected strokes, a pen-up state reduced the desired force and moved the tool away from the wall to avoid unintended connecting lines. The brush front was represented by the end-effector position plus a constant tool-radius offset. Fig. 3 shows the Piper manipulator executing the prescribed UCLA trajectory in MuJoCo.

B. Directionally Decomposed Control Architecture

The controller separates the contact task into three coordinated objectives. The wall-normal x channel regulates contact through a compliant force-to-motion mapping. The y - z channel supplies the nominal writing trajectory, and the orientation channel keeps the tool axis aligned with the wall normal. The resulting Cartesian target is converted to six joint-position references through damped inverse kinematics.

Let F_{des} denote the desired normal force and F_{meas} the estimated force. The outer-loop compliant displacement $e = x_{\text{imp}} - x_{\text{ref}}$ is governed by the virtual mass-damping-stiffness dynamics

$$M\ddot{e} + B\dot{e} + Ke = F_{\text{des}} - F_{\text{meas}}, \quad (1)$$

where M , B , and K are virtual controller parameters rather than physical properties of the robot. The implemented values were

$$M = 1.6 \text{ kg}, \quad B = 145 \text{ N s m}^{-1}, \quad K = 560 \text{ N m}^{-1}. \quad (2)$$

At each time step, Eq. (1) is rearranged as

$$\ddot{e} = \frac{F_{\text{des}} - F_{\text{meas}} - B\dot{e} - Ke}{M}, \quad (3)$$

and numerically integrated to obtain virtual velocity and position. Velocity, displacement, and command-rate limits are applied after integration to improve safety and numerical stability. The x command is then combined with the

Algorithm 1 Simulation indirect force controller

- 1: Generate nominal letter position $\mathbf{x}_{\text{nom}}(t)$ in the wall plane.
 - 2: Estimate F_{meas} using Eq. (6).
 - 3: Integrate Eq. (1) to update the wall-normal compliant state.
 - 4: Rate-limit and clamp the normal command to obtain x_{cmd} .
 - 5: Set $\mathbf{p}_{\text{cmd}} = [x_{\text{cmd}}, y_{\text{traj}}, z_{\text{traj}}]^T$.
 - 6: Solve damped inverse kinematics and send q_{cmd} to the simulated robot.
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prescribed tangential trajectory to form

$$\mathbf{p}_{\text{cmd}} = [x_{\text{cmd}}, y_{\text{traj}}, z_{\text{traj}}]^T. \quad (4)$$

A tool-axis orientation error is added to the Cartesian task, and the sixth joint is held near a reference value to suppress unnecessary tool roll. The damped pseudoinverse is used to reduce amplification of joint motion near unfavorable configurations:

$$J^\# = J^T (JJ^T + \lambda^2 I)^{-1}. \quad (5)$$

The resulting joint increment includes posture regularization so that the arm maintains a consistent configuration while preserving the primary Cartesian task. In the reported simulation, a quasi-static inner servo applied the inverse-kinematics solution nearly directly to the joint states. This assumption isolates the outer-loop contact behavior but does not reproduce the finite bandwidth of the physical Piper joint servos.

C. Contact-Force Estimation and Feedback

The simulation did not use a physical six-axis force/torque sensor. Instead, normal contact force was estimated from penetration of the tool front into the wall and relative closing velocity. The unilateral spring-damper model was

$$F_{\text{meas}} = \max(0, K_c \delta + B_c v_{\text{rel}}), \quad (6)$$

where δ is normal penetration, v_{rel} is relative normal velocity, $K_c = 3000 \text{ N m}^{-1}$ is contact stiffness, and $B_c = 60 \text{ N s m}^{-1}$ is contact damping. The maximum operator prevents negative tensile force and represents nonadhesive contact: the wall can push the tool but cannot pull it after separation.

The wall pose is known to the force estimator because it is needed to calculate penetration and relative velocity. However, the instantaneous wall position is not directly copied into the x -position reference. When the wall retreats, penetration and estimated force decrease first. The resulting positive force error then drives Eq. (1), causing the tool command to move toward the wall. Thus, wall following is feedback driven, although the feedback itself remains model dependent.

D. Hardware Controller: Joint-Level MIT Impedance

The first real-robot approach attempted to add a virtual spring on top of the robot's Cartesian position command interface. This was not sufficiently compliant because the on-board high-stiffness position servo treated the commanded–actual position gap as a tracking error to eliminate. To achieve safer physical interaction, the final implementation uses the Piper low-level MIT joint interface through the Piper SDK [7]:

$$\tau_i \approx K_{p,i} (q_{\text{ref},i} - q_i) + K_{d,i} (-\dot{q}_i), \quad (7)$$

where $q_{\text{ref},i}$ is obtained from Pinocchio inverse kinematics and each joint behaves approximately like a virtual spring–damper. Pinocchio is an open-source library for rigid-body kinematics and dynamics algorithms [9]. In the real controller, the gripper joints are locked in the reduced kinematic model, leaving six controllable joints.

The real-world controller uses the following nominal unscaled gains from the final script:

$$K_p = [10, 8, 8, 8, 6, 6], \quad (8)$$

$$K_d = [0.8, 0.6, 0.6, 0.6, 0.5, 0.5]. \quad (9)$$

The single stiffness scale s modifies both stiffness and damping:

$$K_p^{\text{eff}} = sK_p, \quad K_d^{\text{eff}} = \sqrt{s}K_d. \quad (10)$$

The recommended setting is $s = 0.3$, which gives

$$K_p^{\text{eff}} = [3.0, 2.4, 2.4, 2.4, 1.8, 1.8] \quad (11)$$

and

$$K_d^{\text{eff}} \approx [0.44, 0.33, 0.33, 0.33, 0.27, 0.27]. \quad (12)$$

The command rate is 200 Hz, the ramp time is 2 s, the letter speed is 0.02 m s^{-1} , and the maximum per-joint position error is 0.35 rad, approximately 20° . The error limit bounds the approximate proportional torque contribution:

$$|\tau_{p,i}| \leq K_{p,i}^{\text{eff}} e_{q,\text{max}}. \quad (13)$$

This bound is a safety-oriented design choice: increasing stiffness improves tracking but increases the risk of excessive contact force and end-effector damage.

E. Explicit Assumptions

The simulation and hardware controllers were designed under the following assumptions:

- 1) Robot kinematic parameters are exact, and the wall is planar and rigid except for its prescribed translational motion in simulation.
- 2) The tool-radius offset is constant; bristle deformation, eraser deformation, paint dynamics, and color-removal mechanics are not modeled.
- 3) The simulation contact stiffness and damping are constant, linear, and known to the force estimator.
- 4) Sensor noise, quantization, network latency, and computation delay are neglected in simulation.

TABLE II: Simulation contact-force metrics for the recommended gains.

Metric	Value	Interpretation
Task duration	38.99 s	Reduced from approximately 96 s
Desired pen-down force	10.00 N	Reference
Mean pen-down force	10.99 N	0.99 N positive bias
Force RMSE	1.45 N	Pen-down samples
Peak pen-down force	13.86 N	Initial contact transient

- 5) The quasi-static inner joint-position servo in simulation tracks inverse-kinematics commands without meaningful delay or error.
- 6) The real robot does not directly measure normal force; real-world contact regulation is therefore indirect and based on joint compliance.

V. RESULTS

A. Simulation Force Regulation

Figure 4 shows the response after the painting trajectory was conservatively accelerated. The complete task duration was reduced from approximately 96 s to 38.99 s. Over pen-down samples, the mean force was 10.99 N, the force-error RMSE was 1.45 N, and the peak force was 13.86 N. The largest peak occurred during initial contact establishment. The response remained strongly damped and did not exhibit sustained oscillation.

During intentional pen-up intervals, the desired force was set to zero and the estimated force correspondingly fell to zero. The approximately 0.99 N positive difference between the mean pen-down force and the 10 N target indicates a remaining steady force bias. This offset is consistent with the spring term in the virtual dynamics and the absence of integral compensation.

B. Wall Following and Contact Maintenance

As shown in Fig. 5, the prescribed wall motion included both forward and backward displacement to emulate a change in surface height. During the 6.00 mm retreat event, the brush front followed 4.972 mm, producing a displacement-following ratio of

$$\text{Following ratio} = \frac{\Delta x_{\text{brush}}}{\Delta x_{\text{wall}}} \times 100\% = 82.86\%. \quad (14)$$

The contact gap remained negative for 100 % of the retreat samples, satisfying the primary contact-maintenance requirement. A negative gap denotes finite tool compression rather than separation. Positive gap excursions outside the retreat interval correspond to intentional pen-up transitions and are not classified as controller failure.

During the retreat interval, the mean force was 12.05 N and the force RMSE relative to the 10 N target was 2.27 N. Continuous contact was therefore achieved at the cost of a temporary positive force bias. The result demonstrates that the outer compliant loop can respond to a retreating surface through force feedback, even though it does not reproduce the full wall displacement.

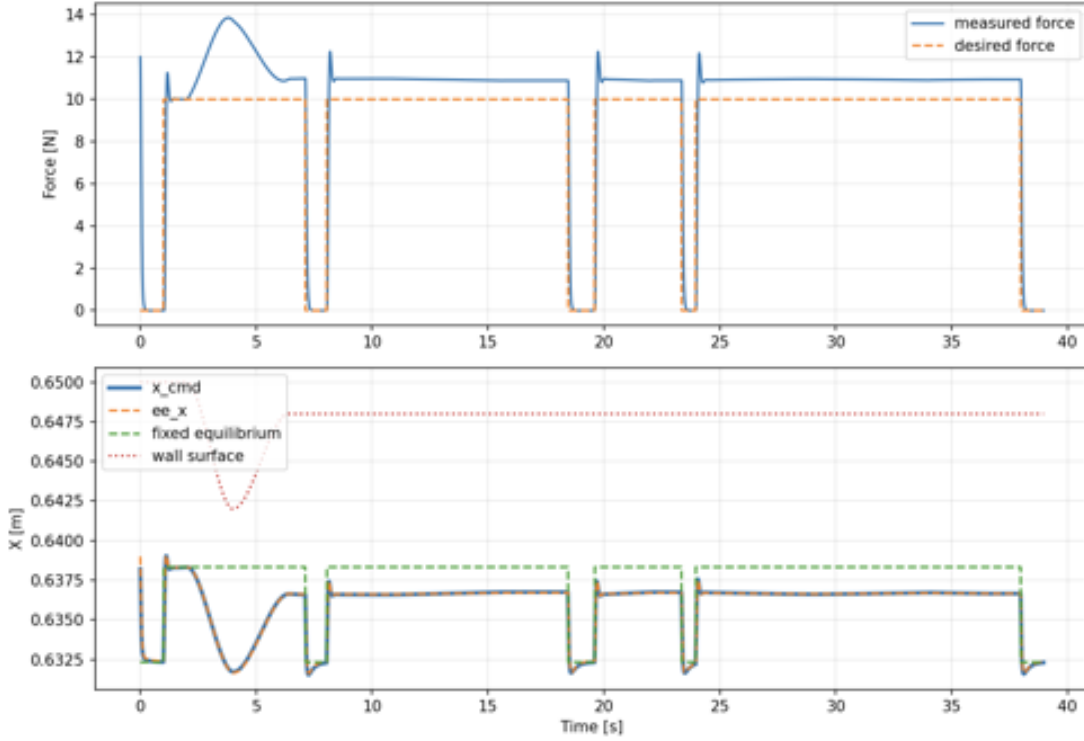


Fig. 4: Contact-force response and wall-normal controller states. The dashed orange line denotes the desired force, and the lower plot shows commanded, measured, equilibrium, and wall-surface positions.

TABLE III: Simulation path, contact, and orientation summary.

Metric	Measured	Requirement	Status
Mean pen-down e_{yz}	0.497 mm	< 1 mm	Pass
Maximum pen-down e_{yz}	1.013 mm	< 2 mm	Pass
Maximum brush-axis drift	0.062°	$< 1^\circ$	Pass
Contact during retreat	100 %	100 %	Pass
Wall-following ratio	82.86 %	Response metric	Reported

C. Tangential Path and Orientation Performance

Figure 6 shows that the nominal and actual UCLA trajectories remain closely aligned. For pen-down samples, the mean Euclidean y - z error was 0.497 mm and the maximum was 1.013 mm. Both values satisfy the specified requirements of less than 1 mm mean error and less than 2 mm maximum error. The tangential error is

$$e_{yz} = \sqrt{(y - y_{\text{des}})^2 + (z - z_{\text{des}})^2}. \quad (15)$$

The full time history contains transition peaks of approximately 2.4 mm. These peaks occur mainly during pen-up travel and abrupt changes between path segments; they are excluded from the reported pen-down maximum because they do not represent the accuracy of painted strokes. The maximum brush-axis drift was 0.062°, substantially below the 1° requirement.

D. Hardware Reverse-Painting Result

The hardware test executed the four letters “UCLA.” The tool did not deposit fresh paint; instead, the wall was colored in advance, and the end-effector eraser removed color along the commanded strokes. This reverse-painting method made the final path visible while preserving the same contact-control challenge: the robot had to maintain sufficient normal pressure for erasing without becoming overly stiff against the wall.

The final hardware run used the joint-level MIT impedance controller with a 200 Hz control rate, a 2 s ramp time, a 0.02 ms^{-1} letter speed, and a 0.35 rad joint-error limit. The stiffness scale was selected as $s = 0.3$. This value was a practical compromise: $s = 0.1$ was too soft and produced weak or distorted contact because the tool did not press consistently enough against the colored wall; $s = 0.3$ provided sufficient contact and legibility while remaining compliant; $s \geq 0.5$ made the arm too stiff, increasing the risk of surface damage, tool damage, and unsafe contact forces. Because no force/torque sensor was used in the hardware setup, these results are evaluated qualitatively through visible letter formation, contact stability, and safety.

VI. DISCUSSION

A. Effects of Controller Gains

The virtual mass, damping, and stiffness determine the trade-off among responsiveness, force stability, and geometric compliance. The mass M is a virtual controller parameter,

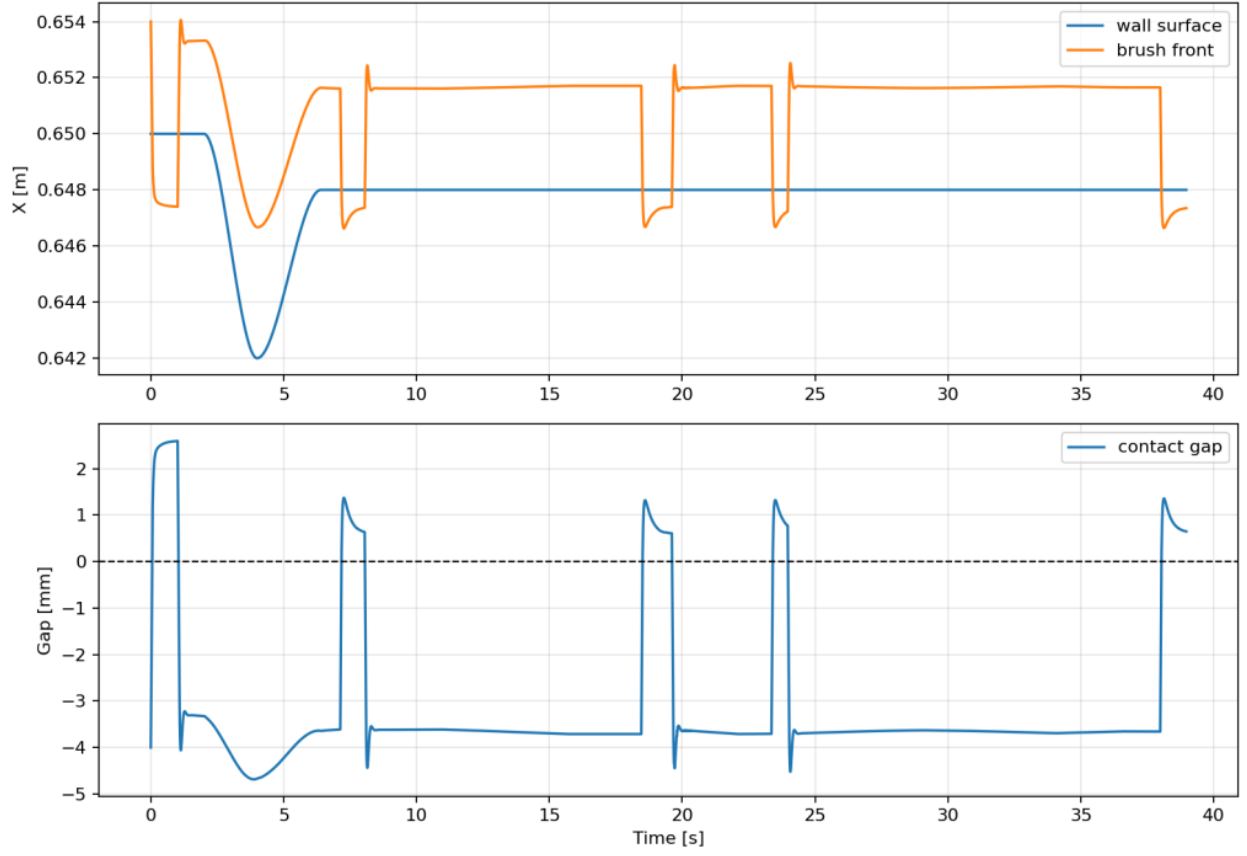


Fig. 5: Wall position, brush-front response, and contact gap. The zero-gap dashed line separates compression from geometric separation.

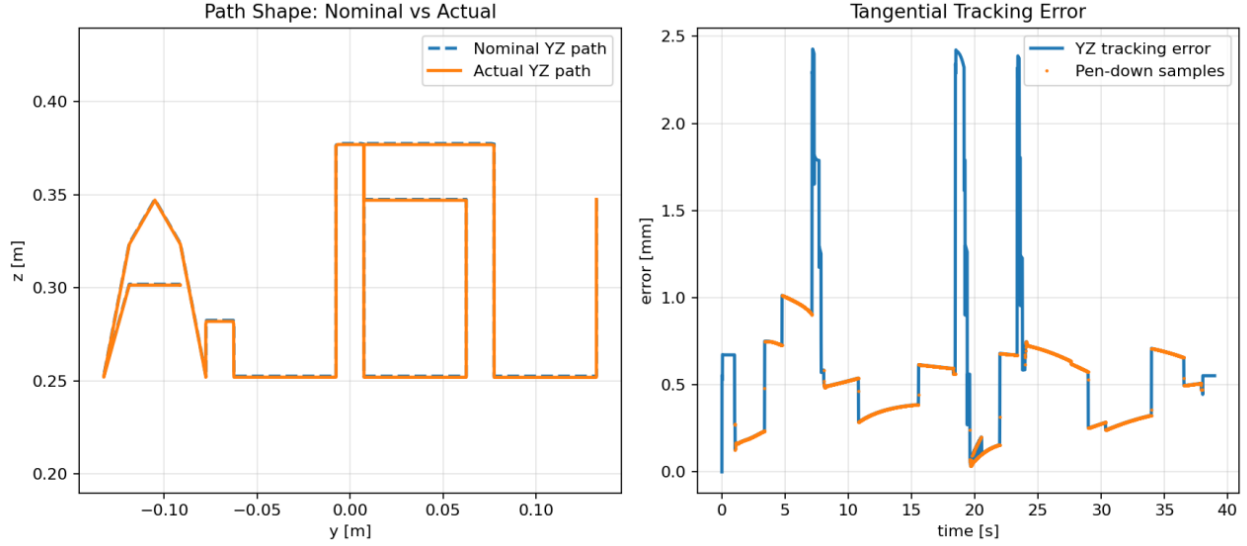


Fig. 6: Nominal and actual y - z paths on the wall plane and Euclidean tangential tracking error. Pen-down samples are shown as individual markers.

not the physical mass of the robot. Increasing M reduces the acceleration produced by a given force error, yielding smoother but slower x -direction correction. Decreasing M

improves responsiveness but increases sensitivity to discrete-time noise and transient oscillation.

Increasing damping B suppresses overshoot and ringing

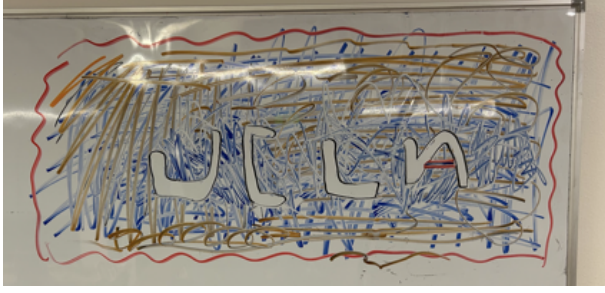


Fig. 7: Final hardware reverse-painting result. The colored whiteboard background was erased along the commanded path to form the four letters “UCLA.”

TABLE IV: Hardware controller settings and stiffness observations.

Quantity	Value or observation
Control rate	200 Hz
Ramp time	2 s
Letter speed	0.02 m s^{-1}
Joint error limit	$0.35 \text{ rad} \approx 20^\circ$
Letters tested	Full “UCLA”
Hardware process	Reverse painting by erasing pre-colored wall
Too soft	$s = 0.1$
Recommended	$s = 0.3$
Too stiff	$s \geq 0.5$

during contact establishment. Excessive damping, however, creates lag and reduces the tool’s ability to follow rapid wall retreat. Decreasing B improves response speed but can produce force peaks, repeated contact switching, and unstable-looking motion.

Increasing stiffness K pulls the compliant state more strongly toward the fixed equilibrium and reduces normal displacement. This can reduce drift, but it also weakens compliance and makes the response resemble rigid position control. Decreasing K improves wall-following ability but allows larger x excursions and can make force-error rejection slower.

The final simulation setting, $M = 1.6 \text{ kg}$, $B = 145 \text{ N s m}^{-1}$, and $K = 560 \text{ N m}^{-1}$, was intentionally highly damped. It prevented sustained oscillation, preserved contact during the complete retreat, and maintained submillimeter mean pen-down error. The remaining startup overshoot and steady force bias suggest that future improvement should focus on compliant-state initialization, bias compensation, and testing with nonideal actuator dynamics rather than simply increasing stiffness.

The hardware stiffness scale s was the most important real-world tuning parameter. With $s = 0.1$, the arm was too compliant to reliably remove color from the wall; the eraser could deflect away from the desired contact state and the letters became weak or inconsistent. With $s \geq 0.5$, the arm behaved more like a stiff position servo, which improved geometric tracking but made contact less forgiving and increased the risk of damaging the wall, eraser, or arm. The selected value $s = 0.3$ is therefore recommended for the

TABLE V: Observed gain effects.

Gain	Effect when increased	Main trade-off
Virtual mass M	Smoother acceleration	Slower wall following
Damping B	Less overshoot and oscillation	Greater lag
Stiffness K	Less normal drift	Lower compliance
Hardware scale s	Better letter tracking/pressure	Higher damage risk

current reverse-painting setup because it provides adequate contact pressure while preserving compliance.

B. Selection and Recommendation of Contact-Control Strategy

One major challenge was selecting an appropriate contact-control strategy for a position-actuated robot. Direct torque-level Cartesian impedance control would convert Cartesian motion error into force or torque commands, but the available simulation and hardware interfaces did not both support a reliable, identical low-level torque-control loop. Attempts to reproduce torque-like behavior through additional correction, blending, and filtering increased delay and degraded the written trajectory.

The recommended simulation controller is therefore an outer-loop admittance controller with an inner joint-position loop. The measured force error is passed through the virtual mass-damping-stiffness dynamics to generate an x -direction position correction, which is then combined with the y - z path and mapped to joint positions through damped inverse kinematics. This architecture produced readable letters, responsive wall following, and low oscillation for the available interface.

The recommended hardware controller is the joint-level MIT impedance controller with $s = 0.3$, 200 Hz command rate, 2 s ramp time, and 0.35 rad joint-error limit. It should be described as an indirect impedance implementation rather than direct force control, because hardware normal force is not directly measured. For a future robot with accurate external-wrench sensing and reliable torque control, a genuine torque-level Cartesian impedance or hybrid force/motion controller would be more appropriate.

C. Capabilities, Limitations, and Project Challenges

The primary capability of the controller is directional task separation. Normal compliance responds to wall motion without directly changing the geometry of the y - z letter path. The orientation objective prevents the tool from rotating away from the wall, and damped inverse kinematics reduces numerical instability near unfavorable configurations. The reported simulation satisfies all stated contact, path, and orientation requirements.

The most important development challenges were maintaining stable contact while preserving letter geometry, preventing unnecessary wrist motion over a long trajectory, and interpreting impedance and admittance terminology correctly for a position-controlled robot. The final task decomposition addressed shape distortion by confining compliance mainly to x while retaining explicit y - z and orientation objectives.

TABLE VI: Representative failure modes.

Failure condition	Cause	Likely outcome
Wall retreats too far or too quickly	Displacement, rate, or bandwidth limits exceeded	Positive gap and contact loss
Wall moves inward too quickly	Controller cannot retreat before compression grows	Large overshoot force
Damping too low	Insufficient energy dissipation	Oscillatory force and repeated contact switching
Stiffness too high	Compliance suppressed	Rigid behavior and poor retreat following
IK near singularity	Pseudoinverse amplification	Large or sudden joint commands
Real hardware s too low	Joint compliance too high	Weak erasing and distorted letters
Real hardware s too high	Joint compliance too low	Damage risk and unsafe contact

An additional hardware challenge was the absence of direct force sensing: the system could demonstrate compliant reverse painting, but it could not measure hardware force RMSE in the same way as simulation.

D. Response to Presentation Feedback

The presentation feedback form in Appendix identified four report-level questions that shaped the final discussion. First, the whiteboard was moved in the hardware demonstration because the reachable workspace of the Piper arm was limited for the desired letter size and contact setup from a fixed base pose. This was a practical laboratory workaround rather than a requirement of the control strategy. In a more integrated system, the mobile base could translate the manipulator relative to the wall, allowing the same wall-contact controller to cover a larger writing area without manually moving the whiteboard.

Second, the position plots use meters because the simulation state, wall motion, and virtual trajectory commands are all represented in SI units. The absolute wall and tool displacements are on the centimeter scale, while tracking and contact-gap metrics are reported in millimeters for readability. In the hardware demonstration, the virtual contact target was placed approximately 10 cm from the initial approach point to the final commanded point, with the physical wall generally located between those two positions. This behind-surface virtual target allowed the joint-space impedance controller to generate a sustained normal contact tendency without commanding a rigid penetration of the wall.

Third, the 10 N simulated normal-force target was intentionally conservative for maintaining visible contact and was also motivated by possible rolling-paint applications. However, excessive normal pressure increases tangential friction between the tool and the surface, which can degrade letter tracking, increase actuator effort, and risk damaging the wall or end effector. For the eraser-based reverse-painting hardware task, lower force or adaptive force/stiffness scheduling would likely improve tracking as long as the pressure remains high enough for consistent color removal.

Fourth, joint-space gain tuning was manageable for the final hardware controller because the main tuning variable was a global stiffness scale s after nominal joint gains were selected. Empirically, $s = 0.3$ gave a useful compromise between contact pressure and safety. Operational-space control remains an attractive future direction because it would allow more direct tuning of wall-normal and wall-tangential stiffness. The project did not fully adopt it because the available hardware setup did not provide a reliable measured wrench and torque-level Cartesian control interface for the final deadline.

VII. FUTURE DIRECTIONS

Hardware experiments should compare the recommended joint-impedance architecture with rigid position control using repeated trials. A first improvement is to add a force/torque sensor or calibrated current-based contact estimator so the real system can regulate F_{meas} directly rather than relying only on compliance. A second improvement is to implement adaptive stiffness so that the robot can use low stiffness during contact establishment and higher stiffness only when contact is safe and stable. A third improvement is to evaluate actual painting or erasing quality with quantitative image metrics, such as stroke coverage, letter-shape error, and color-removal uniformity.

Additional future work includes testing curved or textured surfaces, improving the custom roller/eraser CAD so that it passively accommodates wall-angle errors, and validating the same metrics used in simulation—force RMSE, contact-maintenance ratio, wall-following displacement, y - z error, and tool-axis drift—on real hardware with appropriate sensors.

VIII. CONCLUSION

A directionally decomposed compliant controller was developed for a Piper manipulator painting on a moving wall. The wall's forward and backward motion represented local surface-height variation. The simulation controller used an outer-loop admittance model in the wall-normal direction, explicit y - z path tracking, tool-axis orientation control, damped inverse kinematics, and an idealized joint-position inner loop.

After conservative trajectory acceleration, the complete simulated task was executed in 38.99 s. The controller maintained contact throughout a 6.00 mm retreat, with a brush-following displacement of 4.972 mm and a following ratio of 82.86 %. During pen-down motion, the mean force was 10.99 N, the force RMSE was 1.45 N, and the peak force was 13.86 N. The mean and maximum pen-down y - z errors were 0.497 mm and 1.013 mm, and the maximum tool-axis drift was 0.062°.

All declared simulation requirements were satisfied. However, the force was estimated from a known contact model and the inner joint-position loop was idealized. The results therefore demonstrate high-level feasibility and controller-design insight rather than guaranteed hardware performance.

On the real robot, reverse painting was demonstrated by using an end-effector eraser to remove color from a pre-colored wall and write the four letters “UCLA.” The recommended real-world stiffness scale is $s = 0.3$, which balances letter legibility and safe compliance.

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APPENDIX

TABLE VII: Itemized team member contributions.

Member	Contributions
Ruohan He	Completed algorithm validation in simulation and participated in final report writing.
Shixuan Sun	Built the simulation environment and supported simulation setup, testing, and validation.
Jiawen Wang	Built the real hardware platform, improved the real-world MIT impedance algorithm, contributed to the final report, and maintained the GitHub repository.
Xiaoyang Zhao	Designed and fabricated the end-effector using 3D CAD and supported physical integration.

All team members participated in controller-design discussions, including the choice of indirect force-control architecture, simulation-to-hardware transfer, and real-world stiffness selection.

TABLE VIII: Bill of materials used for the hardware demonstration. Costs were not itemized in this report draft.

Item	Qty.	Description	Source
Piper manipulator	1	Six-degree-of-freedom arm	AgileX
Mobile base/platform	1	Robot support platform	AgileX/lab
Custom eraser mount	1	Fabricated end-effector adapter	Team fabricated
Eraser material	1	Contact tip for reverse painting	Lab stock
Whiteboard/wall surface	1	Vertical drawing surface	Lab stock
Control computer/CAN interface	1 set	Real-time command computer and interface	Lab stock

The custom simulation code is `sim.py`. The real-robot MIT impedance controller is `impedance_control_mit.py`. The native files should be submitted in the Google Drive or UCLA Box folder, and the custom code should also be printed into the combined Turnitin PDF according to the course instructions.

TABLE IX: Code artifacts.

File	Purpose
<code>sim.py</code>	MuJoCo simulation, wall model, force estimator, admittance controller, plots, and video rendering.
<code>impedance_control_mit.py</code>	Piper hardware controller using Pinocchio IK and MIT joint impedance commands.
<code>main.tex</code>	IEEE-style report source.
<code>references.bib</code>	Bibliography database.

MAE C163C / C263C Project Presentation Feedback

Team #: 13

Project Title: RoboPainting

TOPIC	COMMENTS
Hard copy of slides?	Yes
Intro	OK
Methods (e.g., concepts applied from class)	- Simulation + hardware implementation - Joint-space impedance control
Results (e.g., video, error assessment) and Discussion	Clever use of a whiteboard for "reverse" painting
Sources cited properly, and on same slide as content?	Yes
Clarity of presentation	OK
Technical competence	OK
Equal presentation effort across all team members?	Yes
Ability to field questions	Why move the whiteboard? Was the workspace too large for the robot arm?
Changes/additions for project report	- Address any questions cited above. - Slide 4: Change y-axis label to be in mm or cm (not m) - By how much (in mm or cm) did you have to command the virtual path to be behind the writing surface? - Do you think you could have gotten better performance if you reduced the contact force? A contact force of 10N is kind of high, but that might be necessary for rolling paint. - How difficult was it for you to tune gains for joint-space control? I would think that operational space control would have been easier for gain tuning.

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Fig. 8: Instructor presentation feedback form for Team 13.